Question 1: Student Class

Create a `Student` class with the following attributes:

- `name`
- `roll_no`
- `course`
- `marks`

Create three student objects and display their details.

Requirements:

- Use a constructor.
- Create a method `display_details()`.
- Calculate and display the percentage assuming marks are out of 500.

Question 2: Bank Account

Create a `BankAccount` class with:

- `account_number`
- `account_holder`
- `balance`

Implement the following methods:

- `deposit(amount)`
- `withdraw(amount)`
- `display_balance()`

Requirements:

- Do not allow withdrawal if the amount is greater than the balance.
- Create at least three bank account objects.
- Demonstrate deposits and withdrawals.

Question 3: Rectangle Calculator

Create a `Rectangle` class with:

- `length`
- `width`

Implement methods to calculate:

- Area
- Perimeter
- Diagonal

Create multiple rectangle objects and display their results.

Question 4: Employee Management

Create an `Employee` class with:

- `employee_id`
- `name`
- `department`
- `salary`

Implement methods to:

- Display employee information.
- Increase salary by a given percentage.
- Calculate annual salary.

Create at least five employees and display their information.

Question 5: Product Inventory

Create a `Product` class with:

- `product_id`
- `name`
- `price`
- `quantity`

Implement methods:

- `display_product()`
- `add_stock(quantity)`
- `sell(quantity)`
- `inventory_value()`

The `sell()` method should not allow selling more items than are available.

Question 6: Car Class

Create a `Car` class with:

- `brand`
- `model`
- `year`
- `speed`

Implement methods:

- `accelerate()`

- `brake()`
- `display_status()`

Rules:

- `accelerate()` increases speed by 10.
- `brake()` decreases speed by 10.
- Speed must never become negative.

Question 7: Temperature Converter

Create a `Temperature` class that stores temperature in Celsius.

Implement methods:

- `to_fahrenheit()`
- `to_kelvin()`
- `display()`

Create objects with different temperatures and display all conversions.

Question 8: Circle Class

Create a `Circle` class with a `radius` attribute.

Implement methods:

- `area()`
- `circumference()`
- `diameter()`

Also create a class variable named `PI`.

Create five circle objects and determine which circle has the largest area.

Question 9: Library Book

Create a `Book` class with:

- `book_id`
- `title`
- `author`
- `price`
- `is_available`

Implement methods:

- `display_book()`
- `borrow_book()`

- `return_book()`

A book cannot be borrowed if it is already borrowed.

Create at least five books and demonstrate the borrowing and returning functionality.

Question 10: Simple Shopping Cart

Create the following classes:

- `Product`
- `ShoppingCart`

`Product` should contain:

- `product_id`
- `name`
- `price`

`ShoppingCart` should allow:

- Adding products
- Removing products
- Displaying cart items
- Calculating total price

Create a small shopping-cart application using these classes.